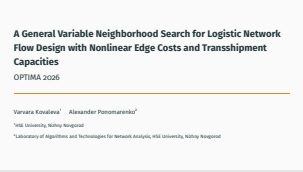
A General Variable Neighborhood Search for Logistic Network Flow Design with Nonlinear Edge Costs and Transshipment Capacities

OPTIMA 2026

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[2]Laboratory of Algorithms and Technologies for Network Analysis, HSE University, Nizhny Novgorod

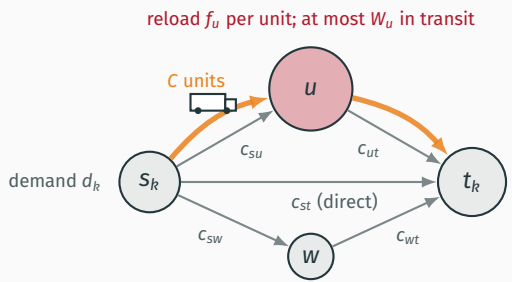

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One breath, then the hook: “Every night, freight between regional sorting centres moves in trucks that are either full or not, and every centre can only reload so much. Those two facts alone break the textbook flow model—this talk is about what to do instead.”

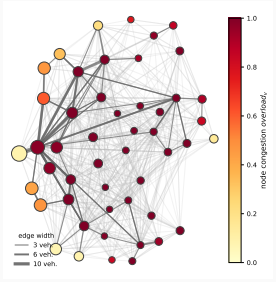
The problem

The problem

Freight between regional sorting centres

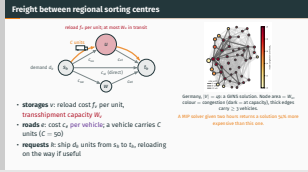


- **storages** v: reload cost f_v per unit, transshipment capacity W_v
- **roads** e: cost c_e per vehicle; a vehicle carries C units ($C = 50$)
- **requests** k: ship d_k units from s_k to t_k, reloading on the way if useful



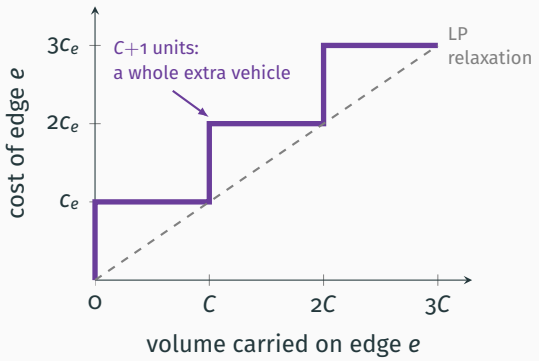
Germany, $|V| = 49$: a GVNS solution. Node area $= W_v$, colour = congestion (dark = at capacity), thick edges carry ≥ 3 vehicles.

A MIP solver given two hours returns a solution 54% more expensive than this one.



Say “trucks” and “reload ceiling”, not “MCND” yet; say out loud that we minimise vehicle movement cost plus reload cost. Point at the map: thick edges are consolidated backbones, dark nodes are saturated storages. The 54% is the ratio 1.544 of Table 2, Germany, MIP-inc. column; the exact budget at this size is 2 h against a 98 min limit for GVNS, of which it used 56 min on average.

Feature 1: a road is billed per truck, not per unit



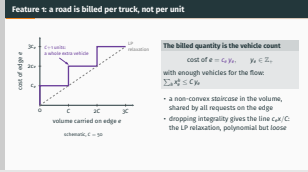
schematic, $C = 50$

The billed quantity is the vehicle count

$$\text{cost of } e = c_e\, y_e, \qquad y_e \in \mathbb{Z}_+$$

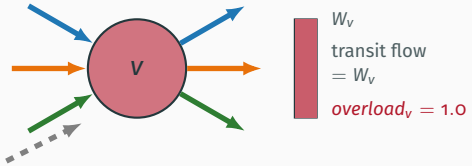
with enough vehicles for the flow:
$\sum_k x_e^k \le C\, y_e$

- a non-convex *staircase* in the volume, shared by all requests on the edge
- dropping integrality gives the line $c_e x / C$: the LP relaxation, polynomial but *loose*



Practitioners recognise the staircase instantly; OR people recognise a step-cost network design problem (Gabrel, Knippel, Minoux 1999; Croxton, Gendron, Magnanti 2003). How loose: the LP solution with vehicles rounded up, our starting solution, costs 1.19× the GVNS solution at $|V| = 10$, 2.8× at $|V| = 50$ and 2.2–3.1× at $|V| \ge 90$ (Table 2).

Feature 2: every storage has a reload ceiling



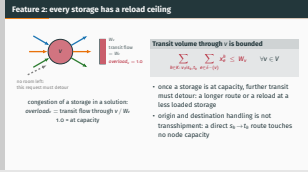
no room left:
this request must detour

congestion of a storage in a solution:
$overload_v$ = transit flow through v / W_v
1.0 = at capacity

Transit volume through v is bounded

$$\sum_{k \in K:\, v \neq s_k, t_k} \; \sum_{e \in \delta^-(v)} x_e^k \;\leq\; W_v \qquad \forall v \in V$$

- once a storage is at capacity, further transit must detour: a longer route or a reload at a less loaded storage
- origin and destination handling is not transshipment: a direct $s_k \to t_k$ route touches no node capacity



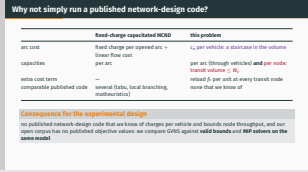
Node throughput bounds are familiar from hub location (Alumur & Kara 2008), not from the MCND literature. $overload_v$ returns twice: it colours the Germany map and it drives neighbourhood N_3.

Why not simply run a published network-design code?

	fixed-charge capacitated MCND	this problem
arc cost	fixed charge per opened arc + linear flow cost	c_e per vehicle: a staircase in the volume
capacities	per arc	per arc (through vehicles) **and** per node: transit volume $\leq W_v$
extra cost term	—	reload f_v per unit at every transit node
comparable published code	several (tabu, local branching, matheuristics)	none that we know of

Consequence for the experimental design

no published network-design code that we know of charges per vehicle and bounds node throughput, and our open corpus has no published objective values: we compare GVNS against **valid bounds** and **MIP solvers on the same model**



Skip in a 20-minute slot and say the consequence on the yardstick slide. Literature, if asked (also on the backup slide "Where it sits"): capacitated MCND – Gendron, Crainic, Frangioni 1999; Crainic 2000; column generation / branch-and-price – Barnhart et al. 1998; cycle-based tabu – Ghamlouche, Crainic, Gendreau 2003; local branching – Rodríguez-Martín & Salazar 2010; MILP-in-the-loop matheuristics – Hewitt, Nemhauser, Savelsbergh 2010; survey – Crainic & Gendreau 2021. Step costs – Gabrel, Knippel, Minoux 1999; Croxton, Gendron, Magnanti 2003. Node capacity – hub location, Alumur & Kara 2008. VNS – Mladenović & Hansen 1997; Hansen et al. 2017; Hansen & Mladenović 2019; multi-start VNS for single-path MCF – Masri et al. 2015; GVNS on CVRP – Amous et al. 2017. Our own real-life logistic heuristics: truck-and-trailer routing with time windows and split deliveries (Batsyn & Ponomarenko 2014, 2015).

Model

Model

The model: an arc-based MILP

$x_e^k \in \mathbb{R}_+$: flow of request k on edge e; $y_e \in \mathbb{Z}_+$: vehicles on e; $|K| \cdot |E| + |E|$ variables.

$$\min \sum_{e \in E} c_e\, y_e \;+\; \sum_{k \in K} \sum_{\substack{v \in V \\ v \neq s_k, t_k}} \sum_{e \in \delta^-(v)} f_v\, x_e^k \qquad \text{vehicles + reloads}$$

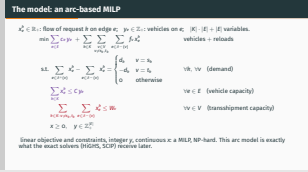
$$\text{s.t.} \sum_{e \in \delta^+(v)} x_e^k - \sum_{e \in \delta^-(v)} x_e^k = \begin{cases} d_k & v = s_k \\ -d_k & v = t_k \\ 0 & \text{otherwise} \end{cases} \qquad \forall k,\ \forall v \quad \text{(demand)}$$

$$\sum_{k \in K} x_e^k \le C\, y_e \qquad \forall e \in E \quad \text{(vehicle capacity)}$$

$$\sum_{k \in K:\, v \neq s_k, t_k} \sum_{e \in \delta^-(v)} x_e^k \le W_v \qquad \forall v \in V \quad \text{(transshipment capacity)}$$

$$x \ge 0, \quad y \in \mathbb{Z}_+^{|E|}$$

linear objective and constraints, integer y, continuous x: a MILP, NP-hard. This arc model is exactly what the exact solvers (HiGHS, SCIP) receive later.



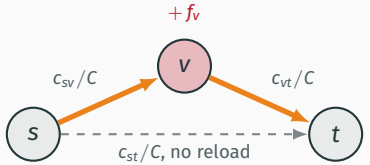
Read the equations once; every term was just seen in pictures. The path-based twin has $\sum_k |\Pi_k| + |E|$ variables. Spoken bridge to the method if the next frame is skipped: relaxing y gives a per-unit path price, travel/C plus one reload per transit node, so the relaxation wants cheap transit-free paths; that is why the start solution is every request on its direct edge, and why N_1 searches with a reload-aware metric.

Path form, and what the LP relaxation tells us

Path form. Flow x_p^k on paths $p \in \Pi_k$ (all $s_k \to t_k$ paths), $\sum_{p \in \Pi_k} x_p^k = d_k$, same y_e; only the paths that carry flow matter.

Relax integrality. The capacity constraint becomes tight, so $y_e = \frac{1}{C} \sum_k \sum_{p \ni e} x_p^k$ and y disappears:

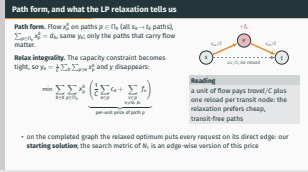
$$\min \sum_{k \in K} \sum_{p \in \Pi_k} x_p^k \underbrace{\left(\frac{1}{C} \sum_{e \in p} c_e + \sum_{\substack{v \in p \\ v \neq s_k, t_k}} f_v \right)}_{\text{per-unit price of path } p}$$



Reading

a unit of flow pays *travel*$/C$ plus one reload per transit node: the relaxation prefers cheap, transit-free paths

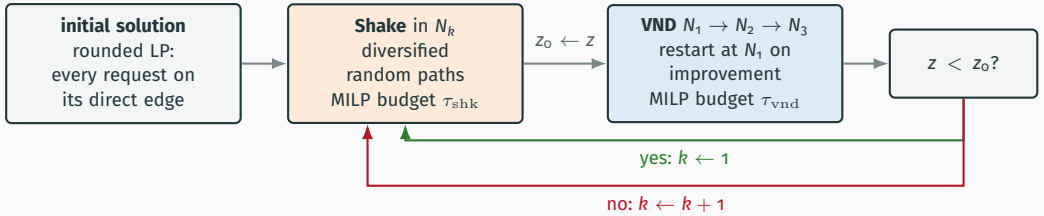
- on the completed graph the relaxed optimum puts every request on its direct edge: our **starting solution**; the search metric of N_1 is an edge-wise version of this price



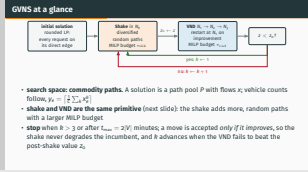
Bridge from model to method; skip in a 20-minute slot (the model frame's note has the spoken version). N_1's metric $c'_{vu} = f_v + f_u + c_{vu}/C$ is edge-wise, so a transit reload is charged on both incident edges; it is a ranking metric for path generation, not the objective—the MILP prices every path exactly.

The GVNS

GVNS at a glance

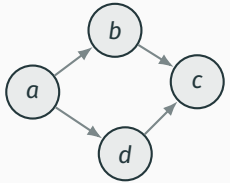


- **search space: commodity paths.** A solution is a path pool P with flows x; vehicle counts follow, $y_e = \left\lceil \frac{1}{C} \sum_k x_e^k \right\rceil$
- **shake and VND are the same primitive** (next slide): the shake adds more, random paths with a larger MILP budget
- **stop** when $k > 3$ or after $t_{\max} = 2|V|$ minutes; a move is accepted *only if it improves*, so the shake never degrades the incumbent, and k advances when the VND fails to beat the post-shake value z_0

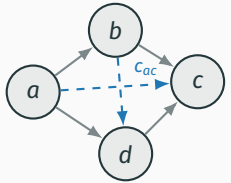


Preprocessing: complete the graph, build the reload-aware graph G', precompute 20 shortest paths per request. Shake vs VND path counts: 5 random pool paths vs 1–3 for N_1; stop at 3 fully loaded paths / 5C free room vs 2 / 1C for N_2; 3 detours vs 1 for N_3. Budgets: $\tau_{\mathrm{shk}} = 10 + |V| \,/\, 100 + 2|V|$ s, $\tau_{\mathrm{vnd}} = 25 \,/\, 70$ s (N_1 / rest; 15 / 50 s at $|V| = 100$), so for N_1 the shake budget is smaller than the VND budget while $|V| \leq 14$. A VNS-school listener will ask whether this is a shake: it is Algorithm 1 called with diversified generation and τ_{shk}; diversification comes from the random paths pushed into the pool and the MILP re-optimisation, and a non-improving shake is discarded. All parameters are on the backup slide “Settings”.

Preprocessing and the starting solution



(a) raw road graph



(b) completed graph

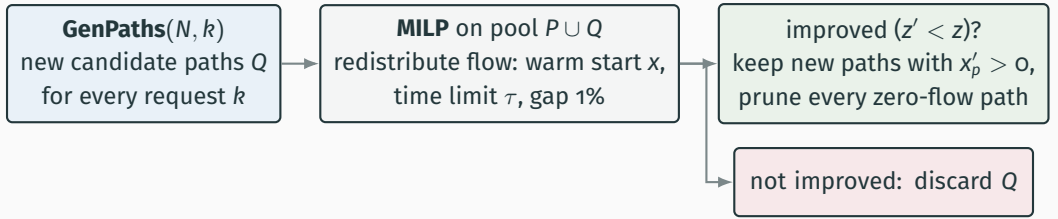
- road graphs are sparse ($a \to c$ needs a stop); on the raw graph a missing origin–destination path can even make an instance infeasible
- **completion**: join every ordered pair by an edge costing the shortest path in G (c_{ac}): the cheapest *stop-free* route, touching no W_v
- the exact solvers get the same completed graph ($|E| = |V|(|V| - 1)$)

Start: the rounded LP

every request on its direct edge (the relaxed optimum), vehicles rounded up: always feasible, and poor (1.19× GVNS at $|V| = 10$, 2.80× at 50, 3.14× on European Russia)

Skip in a 20-minute slot: say “we complete the graph so every request has a direct stop-free edge, and start from the rounded LP”. On the raw graph the node capacities also shrink the feasible region (paper, Sec. 3.2). Many of the trucks on those direct edges leave partly empty—that is what N_2 will fix. The initial MILP that sets up the flows has a 300 s limit.

The move primitive: propose paths, let the MILP redistribute



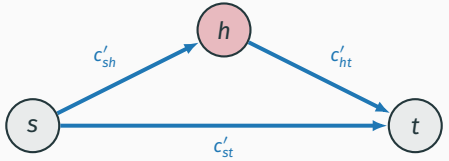
Why paths and not edge flows

per-edge flows are strongly coupled across requests, so a feasible change needs the whole network at once; a path describes one request only, and *adding* a path to a feasible solution never makes it worse

- the three neighbourhoods differ *only* in **GenPaths**; everything else is shared
- pruning zero-flow paths keeps the in-loop MILP small enough for budgets of 25–70 s (VND) and $10+|V|$ to $100+2|V|$ s (shake)
- the neighbourhood proposes, the MILP disposes

This is the matheuristic core; OR people will map it to MILP-in-the-loop schemes (Hewitt et al.). The MILP is HiGHS with a 1% gap target.

N_1: reload-aware shortest paths



Auxiliary complete graph G': per-unit weights

$$c'_{vu} = f_v + f_u + \frac{c_{vu}}{C} \qquad \forall (v,u) \in E$$

travel per unit plus the reloads at the endpoints; plain shortest paths would ignore the reloads

- 20 shortest paths per request under c', precomputed once; new paths come from this pool
- **VND**: $\{1,2,3\}$ new paths per request $\times$ {random, low-cost first}; **Shake**: 5 random pool paths
- ablation: without N_1, $+0.8\%$ (median, synthetic and real alike)

Keep short: N_2 is where the money is. If a sharp listener notes that c' charges a transit reload on both incident edges and also prices origin and destination reloads: c' is a ranking metric for path generation, not the objective; the MILP prices every path exactly, so the double count only changes which 20 paths enter the pool.

N_2: fill the half-empty trucks

edge (u, v): 110 units, $y_{uv} = \lceil 110/50 \rceil = 3$



$load_{uv} = 110 \bmod 50 = 10$:
up to 40 more units here cost **no extra vehicle**
and may free one elsewhere

- shortest paths on G'' run through partially loaded vehicles
- dynamic number of new paths per request: stop after 2 fully loaded paths or accumulated free room $\geq 1C$ (VND); 3 fully loaded paths or $\geq 5C$ (Shake)

Graph G'' favours free room ($\alpha = 0.5$)

$$c''_{vu} = f_v + f_u + \frac{load_{vu}}{C}\left(\frac{c_{vu}}{C}\right)^{\alpha}$$
$$load_e = \left(\sum_k \sum_{p \ni e} x_p^k\right) \bmod C$$

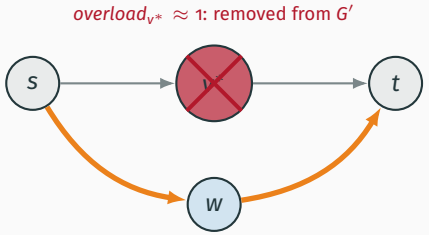
fully loaded edges ($load_e = 0$) keep their G' weight; smaller residual load $\Rightarrow$ lower weight

Ablation: the dominant operator

without N_2: $+3.9\%$ synthetic, $+1.6\%$ real (medians), up to $+23.8\%$ on a single instance

Slow down here. Vehicle consolidation is the main lever on the objective. On the synthetic ablation subset the mean loss is $+7.1\%$ and the worst case $+23.8\%$ (50_nodes_2); on the real subset the mean is $+2.6\%$ and Germany loses 7.6% (components_A.csv); none of these is corpus-wide.

N_3: detour around the saturated storage



in the current solution:

$$overload_v = \frac{1}{W_v} \sum_{k:\, v \neq s_k, t_k} \; \sum_{p \in \Pi_k :\, v \in p} x_p^k$$

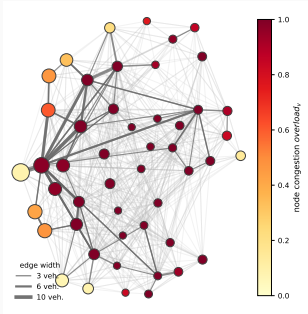
- delete the **single most congested** storage from G' (unless it is the request's own origin or destination), recompute the shortest path: a detour to less loaded storages
- 1 new path per request in VND, 3 in Shake
- ablation: +0.1% / +0.2% median, **inactive when capacities are slack**; +4.0% on Germany, where almost every storage runs at capacity

Say "situational" out loud: the operator is worth carrying only for tight instances—when capacities are slack there is nothing to route around. Point back at the Germany map. If asked why a single node: removing a prefix of $|V|/5$–$|V|/4$ congested storages was tried on all 50 instances $\times$ 10 seeds; it is indistinguishable on 44 of 50 (within across-seed noise) yet forfeits N_3's benefit on the capacity-tight networks, so the surgical version is kept—state the measured outcome, not a mechanism.

Experiments

The benchmark: realistic regional networks, openly available

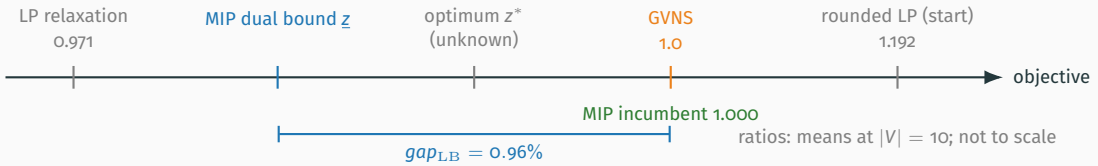
- our open corpus (arXiv:2606.20653; Zenodo, CC-BY-4.0), every instance reproducible from its published seed
- road networks between regional sorting centres, calibrated on OpenStreetMap for **nine European countries, five U.S. states and European Russia**
- real structure: Zipf city sizes, spatial-network topology, centrality-driven W_v, gravity demand; near-planar, large-diameter graphs
- **35 synthetic**: $10 \times |V|$=10, 7×20, 7×30, 5×50, 3×100, 3×150
- **15 real-geometry**: Denmark ($|V| = 11$) to European Russia ($|V| = 90$)



Germany ($|V| = 49$), the solution from the first slide: almost every storage runs at capacity.

The generator is a separate paper; here only what the study needs. Mean degree is 3.575 in the paper. Point at the map: thick edges are consolidation (N_2) made visible, dark nodes are what N_3 works on. The instance's numbers come where they are defined: $gap_{\mathrm{LB}} = 11.0\%$, exact incumbent $1.544\times$, rounded-LP start $2.692\times$, σ over 10 seeds 2.0%; without N_2 +7.6%, without N_3 +4.0%.

Setup and yardsticks

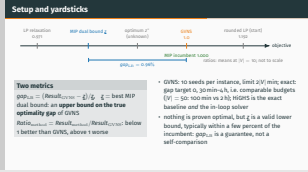


Two metrics

$gap_{\mathrm{LB}} = (Result_{\mathrm{GVNS}} - \underline{z})/\underline{z}$, $\underline{z}$ = best MIP dual bound: an **upper bound on the true optimality gap** of GVNS

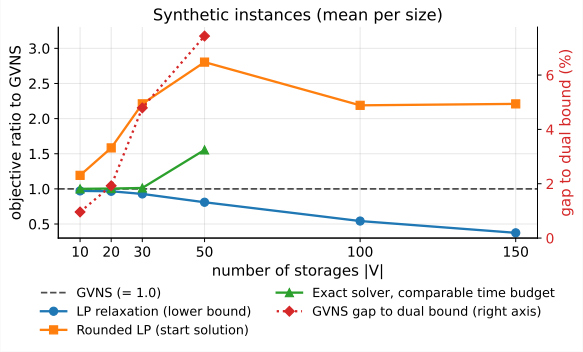
$Ratio_{\mathrm{method}} = Result_{\mathrm{method}}/Result_{\mathrm{GVNS}}$: below 1 better than GVNS, above 1 worse

- GVNS: 10 seeds per instance, limit $2|V|$ min; exact: gap target 0, 30 min–4 h, i.e. comparable budgets ($|V| = 50$: 100 min vs 2 h); HiGHS is the exact baseline *and* the in-loop solver
- nothing is proven optimal, but $\underline{z}$ is a valid lower bound, typically within a few percent of the incumbent: gap_{LB} is a guarantee, not a self-comparison



Hardware: HSE cluster cHARISMa, nodes with 2×AMD EPYC (44 cores, ≥720 GB RAM), each job pinned to 8 cores; HiGHS 1.14.0, SCIP 10.0.2 as cross-check. Exact side in one breath: none of the 50 instances was solved to proven optimality; up to $|V| = 42$ the two solvers' dual bounds agree within 0.6%; SCIP returns nothing usable from $|V| = 45$; the arc model is out of memory at $|V| \geq 100$ (backup "The exact side"). If you accept one number from this talk, make it gap_{LB}: it bounds the distance to the optimum instead of comparing GVNS with itself. The ladder ticks are the $|V| = 10$ means of Table 2; the dual bound sits between the LP and GVNS, and GVNS is 0.96% above it.

Results I: synthetic instances

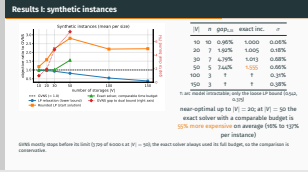


$\lvert V\rvert$	n	gap_{LB}	exact inc.	σ
10	10	0.96%	1.000	0.06%
20	7	1.92%	1.005	0.18%
30	7	4.79%	1.013	0.68%
50	5	7.44%	1.555	0.66%
100	3	†	†	0.31%
150	3	†	†	0.38%

†: arc model intractable; only the loose LP bound (0.542, 0.375)

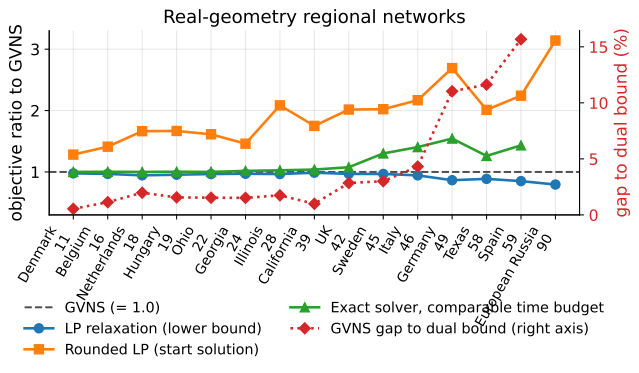
near-optimal up to $|V| = 20$; at $|V| = 50$ the exact solver with a comparable budget is 55% more expensive on average (16% to 137% per instance)

GVNS mostly stops before its limit (3 729 of 6 000 s at $|V| = 50$); the exact solver always used its full budget, so the comparison is conservative.



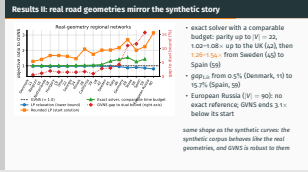
Trace the green curve: flat at 1.0, up to 1.555 at 50, then it stops existing. Spoken: gap_{LB} is about 1–2% up to $|V| \approx 20$, 4.8% at 30, 7.4% at 50, where the LP bound alone would say about 24%; at $|V| \geq 100$ GVNS still ends 2.2× below its own start. Mean run time vs limit: 633/1 200 s (10), 1 952/2 400 (20), 2 611/3 600 (30), 3 729/6 000 (50), 5 983/12 000 (100), 18 236/18 000 (150); the 150-node mean slightly exceeds its limit because Algorithm 2 tests $t_{\max}$ only before each shake and after each full VND sweep, so a shake started just under the limit (MILP budget up to $100 + 2|V| = 400$ s at $|V| = 150$) and the sweep after it run to completion. Per-instance $|V| = 50$ numbers are on the backup slide “What the means hide”.

Results II: real road geometries mirror the synthetic story



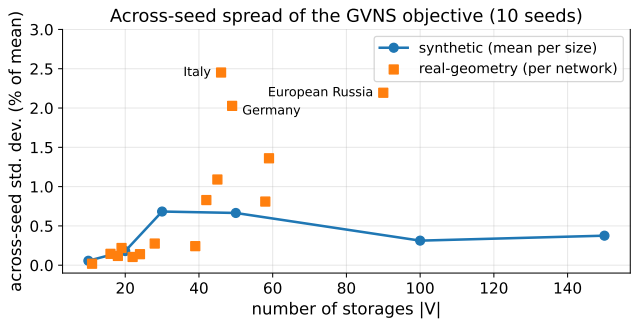
- exact solver with a comparable budget: parity up to $|V| = 22$, 1.02–1.08× up to the UK (42), then 1.26–1.54× from Sweden (45) to Spain (59)
- gap_{LB} from 0.5% (Denmark, 11) to 15.7% (Spain, 59)
- European Russia ($|V| = 90$): no exact reference; GVNS ends 3.1× below its start

same shape as the synthetic curves: the synthetic corpus behaves like the real geometries, and GVNS is robust to them

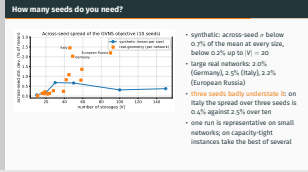


Do not read the table; point at the three medium networks and at the 90-node one. Spoken detail: gap_{LB} about 1–3% up to $|V| = 45$, 4.3% on Italy (46), 11.0% Germany (49), 11.6% Texas (58), 15.7% Spain (59); exact incumbent 1.02–1.04× from Georgia (24) to California (39), 1.077 UK; run time 615 s (Denmark) to 3 986 s (Spain), 5 595 s on European Russia. Full Table 2 rows are in the paper.

How many seeds do you need?

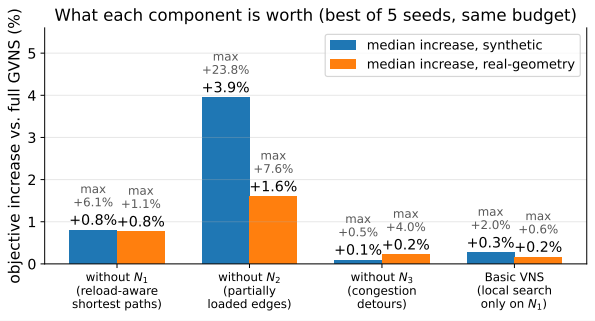


- synthetic: across-seed σ below 0.7% of the mean at every size, below 0.2% up to $|V| = 20$
- large real networks: 2.0% (Germany), 2.5% (Italy), 2.2% (European Russia)
- three seeds badly understate it: on Italy the spread over three seeds is 0.4% against 2.5% over ten
- one run is representative on small networks; on capacity-tight instances take the best of several



Backup for the “is it stable?” question. We moved from 3 to 10 seeds for the headline numbers and 5 for the component study.

What each component buys

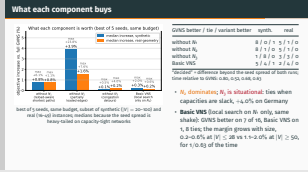


best of 5 seeds, same budget, subset of synthetic ($|V| = 20$–100) and real (16–49) instances; medians because the seed spread is heavy-tailed on capacity-tight networks

GVNS better / tie / variant better	synth.	real
without N_1	8 / 0 / 1	5 / 1 / 0
without N_2	8 / 1 / 0	5 / 1 / 0
without N_3	1 / 8 / 0	3 / 3 / 0
Basic VNS	5 / 4 / 1	2 / 4 / 0

"decided" = difference beyond the seed spread of both runs; time relative to GVNS: 0.80, 0.52, 0.68, 0.63

- N_2 dominates; N_3 is situational: ties when capacities are slack, $+4.0\%$ on Germany
- **Basic VNS** (local search on N_1 only, same shake): GVNS better on 7 of 16, Basic VNS on 1, 8 ties; the margin grows with size, 0.2–0.6% at $|V| \leq 28$ vs 1.1–2.0% at $|V| \geq 50$, for $1/0.63$ of the time



One line: N_2 dominates, the VND matters at scale, N_3 is for tight instances. Table 3's time ratios are the means of the synthetic and real groups (CSV: N_1 0.76/0.84, N_2 0.54/0.50, N_3 0.73/0.63, Basic VNS 0.68/0.58). The CSV's `argmax_instance` names the instance where the variant is *worst*, not best. Single instances can flip sign: the two "variant better" entries in the table are decided (beyond the seed spread of both runs): 50_nodes_2, where dropping N_1 improved the result by 2.1%, and 30_nodes_2, where Basic VNS beat GVNS by 1.3%; Basic VNS also improves Germany by 4.0%, but inside its 4.2% seed spread there, so a tie.

Closing

Closing

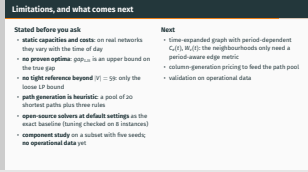
Limitations, and what comes next

Stated before you ask

- **static capacities and costs**: on real networks they vary with the time of day
- **no proven optima**: gap_{LB} is an upper bound on the true gap
- **no tight reference beyond** $|V| = 59$: only the loose LP bound
- **path generation is heuristic**: a pool of 20 shortest paths plus three rules
- **open-source solvers at default settings** as the exact baseline (tuning checked on 8 instances)
- **component study** on a subset with five seeds; **no operational data** yet

Next

- time-expanded graph with period-dependent $C_e(t)$, $W_v(t)$: the neighbourhoods only need a period-aware edge metric
- column-generation pricing to feed the path pool
- validation on operational data

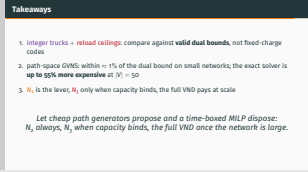


Justifications, spoken: at $|V| = 50$–59 the 7–16% gap cannot be split between heuristic error and bound weakness; beyond $|V| = 59$ the LP bound is 0.795 on European Russia and 0.542 / 0.375 at 100 / 150; HiGHS and SCIP with three presets each moved nothing (backup “The exact side”). The “next” items answer the first, fourth and last limitations.

Takeaways

1. integer trucks + reload ceilings: compare against **valid dual bounds**, not fixed-charge codes
2. path-space GVNS: within $\approx 1\%$ of the dual bound on small networks; the exact solver is **up to 55% more expensive** at $|V| = 50$
3. N_2 is the lever, N_3 only when capacity binds, the full VND pays at scale

Let cheap path generators propose and a time-boxed MILP dispose: N_2 always, N_3 when capacity binds, the full VND once the network is large.



End on the italic sentence; it calls back to “the neighbourhood proposes, the MILP disposes” on the primitive slide.

Thank you

Code, exact-solver harness, scripts that regenerate every table and figure
`github.com/aponom84/Optimal-Logistic-Net`

Instance corpus (CC-BY-4.0)
`doi.org/10.5281/zenodo.20372456`

Contact
`aponomarenko@hse.ru` ·
`varvara_kovaleva03@mail.ru`

Supported by the Basic Research Program at HSE University (HSE-BR-2025-080) and by the HPC facilities of HSE University (cHARISMa).

code

data

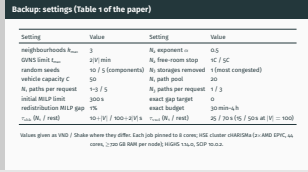


Question map: seeds – “How many seeds”; solver behaviour and tuning – backup “The exact side”; per-instance spread at $|V| = 50$ – backup “What the means hide”; any parameter – backup “Settings”; literature – backup “Where it sits”; the N_3 prefix variant – the N_3 slide’s note.

Backup: settings (Table 1 of the paper)

Setting	Value	Setting	Value
neighbourhoods $k_{\max}$	3	N_2 exponent α	0.5
GVNS limit $t_{\max}$	$2\lvert V\rvert$ min	N_2 free-room stop	1*C* / 5*C*
random seeds	10 / 5 (components)	N_3 storages removed	1 (most congested)
vehicle capacity *C*	50	N_1 path pool	20
N_1 paths per request	1–3 / 5	N_3 paths per request	1 / 3
initial MILP limit	300 s	exact gap target	0
redistribution MILP gap	1%	exact budget	30 min–4 h
τ_{shk} (N_1 / rest)	$10{+}\lvert V\rvert$ / $100{+}2\lvert V\rvert$ s	τ_{vnd} (N_1 / rest)	25 / 70 s (15 / 50 s at $\lvert V\rvert = 100$)

Values given as VND / Shake where they differ. Each job pinned to 8 cores; HSE cluster cHARISMa (2×AMD EPYC, 44 cores, ≥720 GB RAM per node); HiGHS 1.14.0, SCIP 10.0.2.



Backup: the exact side, and is the baseline understated?

	HiGHS 1.14	SCIP 10	arc MILP size
$\lvert V\rvert \le 42$	dual bound + incumbent	dual bound agrees with HiGHS within 0.6% (mean 0.18%); weaker incumbent	$\approx$ 6 000 variables at $\lvert V\rvert = 10$
$\lvert V\rvert = 45$–59	dual bound + incumbent	no feasible solution, no non-trivial bound	
$\lvert V\rvert \ge 90$	millions of variables: at $\lvert V\rvert = 90$ only SCIP was tried and returns no bound and no incumbent; out of memory for both at $\lvert V\rvert \ge 100$		3.7×10^8 at $\lvert V\rvert = 150$

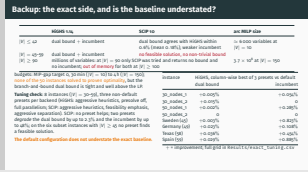
budgets: MIP-gap target 0, 30 min ($\lvert V\rvert = 10$) to 4 h ($\lvert V\rvert = 150$); none of the 50 instances solved to proven optimality, but the branch-and-bound dual bound is tight and well above the LP.

Tuning check: 8 instances ($\lvert V\rvert = 30$–59), three non-default presets per backend (HiGHS: aggressive heuristics, presolve off, full parallelism; SCIP: aggressive heuristics, feasibility emphasis, aggressive separation). SCIP: no preset helps; two presets *degrade* the dual bound by up to 2.3% and the incumbent by up to 48%; on the six subset instances with $\lvert V\rvert \ge 45$ no preset finds a feasible solution.

The default configuration does not understate the exact baseline.

instance	HiGHS, column-wise best of 3 presets vs default dual bound	incumbent
30_nodes_1	+0.005%	+0.054%
30_nodes_2	+0.015%	0
50_nodes_1	+0.002%	+0.285%
50_nodes_2	0	0
Sweden (45)	+0.003%	+0.823%
Germany (49)	+0.023%	+0.108%
Texas (58)	+0.036%	+0.454%
Spain (59)	+0.029%	+0.885%

+ = improvement; full grid in `Results/exact_tuning.csv`

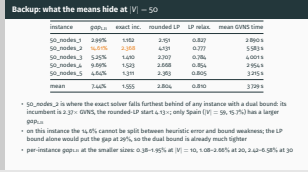


HiGHS is the stronger exact solver here and therefore also the in-loop solver of GVNS. Europe Russia ($\lvert V\rvert = 90$) was attempted with SCIP only (time limit, no bound, no incumbent); the paper reports the arc model as intractable from about $\lvert V\rvert = 90$ and out of memory at $\lvert V\rvert \ge 100$.

Backup: what the means hide at $|V| = 50$

instance	gap_{LB}	exact inc.	rounded LP	LP relax.	mean GVNS time
50_nodes_1	2.99%	1.162	2.151	0.827	2 890 s
50_nodes_2	14.61%	2.368	4.131	0.777	5 583 s
50_nodes_3	5.25%	1.410	2.707	0.784	4 001 s
50_nodes_4	9.69%	1.523	2.668	0.854	2 954 s
50_nodes_5	4.64%	1.311	2.363	0.805	3 215 s
mean	7.44%	1.555	2.804	0.810	3 729 s

- 50_*nodes*_2 is where the exact solver falls furthest behind of any instance with a dual bound: its incumbent is 2.37× GVNS, the rounded-LP start 4.13×; only Spain ($|V| = 59$, 15.7%) has a larger gap_{LB}
- on this instance the 14.6% cannot be split between heuristic error and bound weakness; the LP bound alone would put the gap at 29%, so the dual bound is already much tighter
- per-instance gap_{LB} at the smaller sizes: 0.38–1.95% at $|V| = 10$, 1.08–2.66% at 20, 2.42–6.58% at 30



Backup: where it sits in the literature

Capacitated multicommodity network design

- Gendron, Crainic, Frangioni (1999); Crainic (2000): fixed-charge MCND, NP-hard
- Barnhart et al. (1998): branch-and-price on path formulations
- Ghamlouche, Crainic, Gendreau (2003): cycle-based tabu search
- Rodríguez-Martín, Salazar-González (2010): local branching
- Hewitt, Nemhauser, Savelsbergh (2010): MILP-in-the-loop matheuristics
- Crainic, Gendreau (2021): survey of (meta)heuristics

Step costs and node capacities

- Gabrel, Knippel, Minoux (1999): exact Benders-type method for general step costs
- Croxton, Gendron, Magnanti (2003): MILP models of nonconvex piecewise-linear costs
- Alumur, Kara (2008): hub location with bounded hub throughput

Variable neighbourhood search

- Mladenović, Hansen (1997); Hansen et al. (2017); Hansen, Mladenović (2019): VNS, GVNS
- Karakostas et al. (2019); Mjirda et al. (2016): GVNS vs single-neighbourhood schemes
- Masri, Krichen, Guitouni (2015): multi-start VNS for the single-path multicommodity flow problem
- Amous et al. (2017): VNS for the capacitated VRP

Real-life logistic routing

- Batsyn, Ponomarenko (2014, 2015): truck-and-trailer routing with time windows and split deliveries

Benchmark

- Kovaleva, Ponomarenko (2026): generator and corpus, arXiv:2606.20653, Zenodo 10.5281/zenodo.20372456

